Ideation Phase

Empathize & Discover — Empathy Map Canvas

Date	28 October 2026
Team ID	Ujjawal Rathod (Individual Capstone)
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Empathy Map Canvas: An empathy map captures knowledge about a user's behaviours, goals, and challenges from four perspectives — *Says, Thinks, Does, and Feels* — in relation to the Electric Vehicle Market Performance & Insights project.

Persona 1: Emily Johnson (Market Strategy Director)

SAYS

- "Which EV vehicle categories are driving our highest annual sales volumes right now?"
- "We must identify high-growth segments across model years (2018–2026) before allocating our next R&D and marketing budget."
- "Show me which battery technologies and driving ranges resonate best with satisfied customers."

THINKS

- Investing heavily in commercial EVs (Electric Bus, Electric Truck, Pickup Truck) without verifying demand vs. Two-Wheeler Scooters and Sedans could waste millions.
- With 200 manufacturers competing in a fragmented market, guessing trends without data is too risky.
- Are EV sales climbing steadily year-over-year, or has total volume stabilized across model years?

EMILY JOHNSON — Market Strategy Director (EV Portfolio & Growth)

DOES

- Analyzes multi-year vehicle sales datasets (300,000 unioned records) across 15 vehicle categories and 7 global regions.
- Compares top-performing manufacturers, driving ranges, and average customer ratings side-by-side.
- Uses interactive Tableau filters (Market Region, Battery Type, Model Year) to test strategic scenarios for board presentations.

FEELS

- **Overwhelmed** by raw CSV files containing 300,000 rows, 37 columns, and missing values in sales and ratings.
- **Pressured** to make accurate product-positioning and budget recommendations under tight deadlines.
- **Confident and relieved** when using the interactive 9-visual Tableau dashboard and 4-point Story.

Persona 2: David Chen (Automotive BI Manager)

SAYS	THINKS
• "Who are the Top 10 manufacturers by annual sales, and what is their true share of the global EV market?" • "Which countries lead EV manufacturing production, and which battery chemistries dominate market adoption?" • "Where are the strongest and weakest regional category opportunities across our 7 global markets?"	• Standard Top-10 filters distort market share by making 10 brands add up to 100% instead of comparing against all 200 manufacturers. • Since the top manufacturer (Ampere Works) holds only ~0.59% share, the market is highly fragmented with no single monopoly. • Because regional sales totals (~330M per region) are nearly equal, product category mix matters far more than geography alone.
DAVID CHEN — Automotive BI Manager (Competitive Intelligence)	
DOES	**FEELS**
• Evaluates Top 10 manufacturers using FIXED Level of Detail (LOD) market share calculations in Tableau. • Tracks battery technology adoption (NMC at 33.7% and LFP at 32.0% leading over NCA, Sodium-Ion, and Solid-State). • Cross-filters the Manufacturing Country Map (10 countries led by China, India, Vietnam) and Region × Category Heatmap.	• **Frustrated** when basic spreadsheet charts misrepresent fragmented competitor market shares. • **Analytical and focused** on finding data-backed expansion pockets (such as Crossover demand in Asia Pacific at 29.7M). • **Empowered** by click-to-filter map and donut interactivity on the unified Tableau dashboard.

Persona 3: Dr. Sarah Williams (Sustainable Mobility Researcher)

SAYS	THINKS
• "How does driving range (km) and energy efficiency (Wh/km) compare across all 15 EV vehicle categories?" • "Has average driving range improved across model years (2018–2026) for different battery chemistries?" • "Does customer satisfaction vary significantly by vehicle category or battery technology?"	• Zoomed-in line chart axes (e.g., 319–329 km) create false visual drama over a tiny 5 km fluctuation. • In this dataset, average driving range is driven by vehicle category (Luxury Sedan ~550 km vs. Two-Wheeler Scooter ~105 km), while battery types all average ~324 km. • Customer ratings are remarkably uniform (~4.08 to 4.09 out of 5) across all categories.
DR. SARAH WILLIAMS — Sustainable Mobility Researcher (Battery & Range)	
DOES	**FEELS**
• Examines Average Driving Range by Category colored by Average Efficiency (Wh/km) with a table-wide average reference line. • Audits 2018–2026 battery range trends using a zero-anchored axis (0–400 km) for honest scientific reporting.	• **Skeptical** of exaggerated or poorly scaled visual charts that mislead sustainability policymakers. • **Curious** about why emerging chemistries (Solid-State at ~7%, Sodium-Ion at ~9.9%) show similar average ranges to NMC/LFP in current data.

• Correlates customer satisfaction ratings (0–5 scale) with battery adoption shares for sustainability planning.	• **Satisfied** that the dashboard presents transparent, properly scaled metrics.